

Simulation and Correction of Beam Hardening Artifacts in X-Ray Computed Tomography

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1. Introduction

Computed Tomography (CT) produces high-resolution cross-sectional images using X-ray attenuation and reconstruction algorithms, but image quality is often degraded by artifacts, particularly beam hardening. This effect arises from the polychromatic nature of X-ray sources, where lower-energy photons are preferentially absorbed, causing non-linear attenuation and resulting in distortions such as cupping and streaking.

Beam-hardening effects can be simulated within a full CT pipeline using CTSim, incorporating polychromatic forward projection, sinogram formation, and filtered back-projection reconstruction. Applying polynomial-based correction to the projection data restores linearity and improves image quality, as demonstrated through quantitative comparisons of corrected and uncorrected images.

2. Background and Theory

2.1 CT Imaging Pipeline

The entire process follows a well-defined pipeline that closely mimics the physical acquisition and mathematical reconstruction steps performed by clinical CT scanners. This pipeline can be divided into three main stages:

1. X-ray generation and data acquisition (forward projection).
2. Formation of the sinogram.
3. Image reconstruction.

2.1.1 X-ray Generation and Data Acquisition

X-ray generation in CT begins with a rotating-anode X-ray tube that produces a polychromatic beam (a continuous spectrum of photon energies, typically from ~20 keV up to the peak tube voltage of 80–140 kVp). Unlike an idealized monochromatic source, the emitted spectrum $S(E)$ contains photons of varying energies, with lower-energy photons being more abundant but also more easily absorbed. As the beam traverses the object (the numerical phantom), attenuation follows the polychromatic form of the Beer-Lambert law. For a monochromatic beam of energy E , the transmitted intensity after traveling a path L through material with linear attenuation coefficient $\mu(E, r)$ is:

$$I = I_0 e^{\{-\int \mu(E), ds\}}$$

Low-energy photons are preferentially absorbed by dense tissues, shifting the mean beam energy higher as it penetrates deeper. This makes the effective attenuation coefficient path-length dependent and non-linear, violating the assumptions of standard reconstruction algorithms.

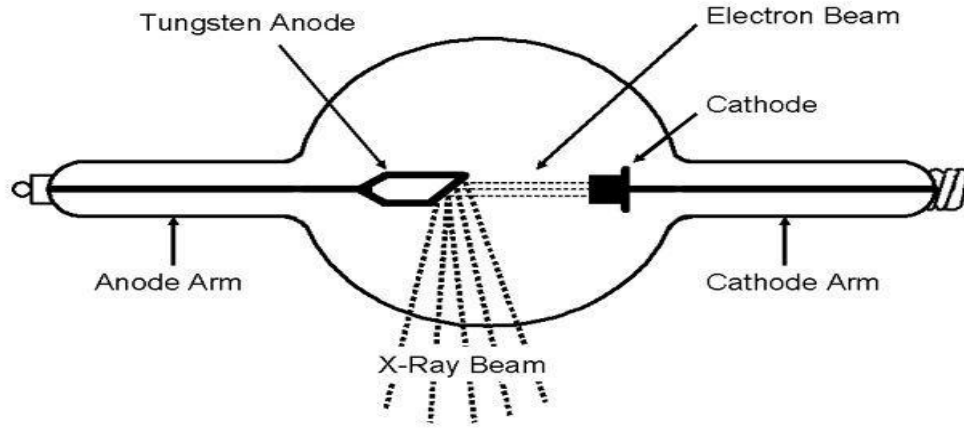


Figure 1: Beer-Lambert law and polychromatic attenuation illustration

In the forward-projection step, rays are cast from the source through the phantom at discrete detector positions and angles. CTSim performs this numerically — it integrates the polychromatic attenuation along each ray using the phantom's material-specific $\mu(E)$. The result is raw intensity data that already incorporates realistic physics, including beam hardening.

2.1.2 Formation of the Sinogram

Once all projections are acquired, they are stacked into a two-dimensional array known as the sinogram. The sinogram organizes the projection data such that the vertical axis represents the projection angle θ (from 0° to 180° or 360°) and the horizontal axis represents the radial detector position t . Each row of the sinogram is one complete projection profile at a fixed angle, and each column traces the sinusoidal path of a single point inside the object as the source-detector pair rotates.

Mathematically, the sinogram $p(\theta, t)$ is the set of all line integrals:

$$p(\theta, t) = \iint \mu(x, y) \cdot \delta(x \cos\theta + y \sin\theta - t) dx dy$$

In CTSim, the sinogram is generated directly from the numerical phantom using parallel-beam geometry with user-specified parameters (number of detectors, views, and samples per detector). The resulting sinogram is a grayscale image where bright regions correspond to high-attenuation paths.

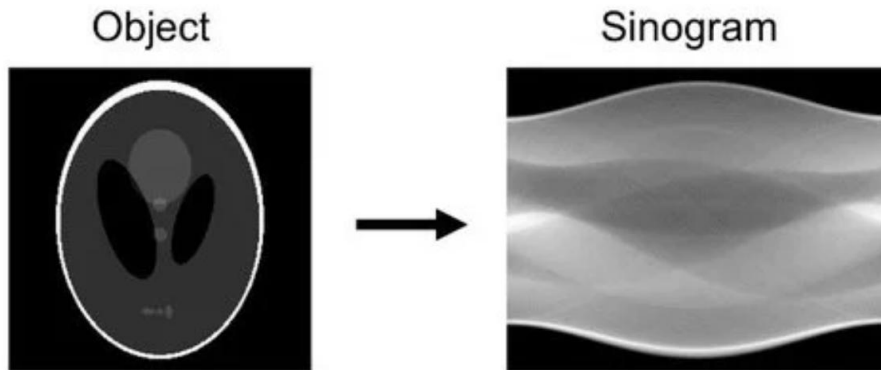


Figure 2: Example sinogram generated from a numerical phantom in CTSim

2.1.3 Image Reconstruction

Image reconstruction recovers the 2-D attenuation map $\mu(x,y)$ from the sinogram. The project employs the classic analytic method Filtered Back-Projection (FBP), which is computationally efficient and the foundation of most clinical CT systems. Simple back-projection smears each projection value back along the original ray paths and sums over all angles. This produces a blurred image because central regions are over-sampled. To correct this, a high-pass "ramp" filter is applied to each projection before back-projection. In the frequency domain, the filter has response $|\rho|$ (where ρ is the radial spatial frequency). The filtered projection $q(\theta,t)$ is:

$$q(\theta, t) = \int p(\theta, \rho) \cdot |\rho| e^{i2\pi\rho t} d\rho$$

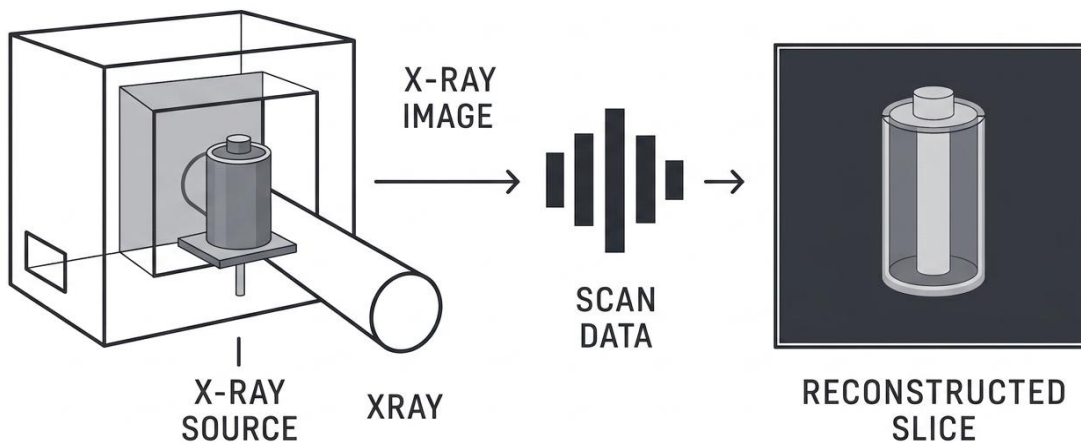


Figure 3: FBP reconstruction pipeline — from sinogram to reconstructed image

FBP assumes the sinogram satisfies the monochromatic Radon transform. When the input sinogram contains polychromatic effects, the reconstruction exhibits cupping and streaking artifacts.

2.2 CT Artifacts

Computed Tomography (CT) artifacts are systematic errors or discrepancies that appear in the reconstructed image but do not correspond to actual anatomical or material properties of the scanned object. These artifacts degrade image quality, reduce diagnostic accuracy, and can lead to misinterpretation of tissue density or contrast. CT artifacts are typically classified by their underlying cause:

- ▶ **Patient-based artifacts:** Caused by patient motion (blurring or streaking), metallic implants/foreign bodies (severe streaking), or high-density contrast agents.
- ▶ **Physics-based artifacts:** Result from the fundamental physics of X-ray interaction with matter.
- ▶ **Scanner-based artifacts:** Due to hardware imperfections such as detector miscalibration (ring artifacts) or tube arcing.
- ▶ **Reconstruction/helical artifacts:** Introduced by the image-reconstruction algorithm itself, especially in multi-slice or helical scanning.

In the context of this project, the focus is on physics-based artifacts, specifically beam hardening. While other artifacts can be minimized through hardware calibration or patient preparation, beam hardening is inherent to the polychromatic nature of clinical X-ray sources and requires explicit software correction in the projection domain.

2.2.1 Beam Hardening Artifact

Beam hardening is a prominent physics-based artifact caused by the polychromatic spectrum of the X-ray beam produced by a clinical or simulated X-ray tube. Unlike the monochromatic beam assumed in standard Filtered Back-Projection (FBP) reconstruction, real X-ray spectra contain a broad range of photon energies. Low-energy photons are preferentially absorbed as the beam passes through dense tissue, increasing the mean energy of the remaining beam. This non-linear attenuation makes the measured projection values path-length dependent, violating the linear Radon-transform assumption.

Because $\mu(E)$ decreases with increasing E , the effective attenuation coefficient drops for longer paths through dense material. This produces two characteristic manifestations in the reconstructed image:

1. **Cupping artifact:** In a uniform-density cylindrical object, the periphery appears brighter (higher attenuation) while the center appears darker (artificially lower attenuation). This creates a cup-shaped profile across the object.
2. **Streaking / dark-band artifact:** Dark streaks or bands appear between two high-density structures because rays passing through both experience greater hardening, leading to inconsistent projection values that FBP cannot reconcile.

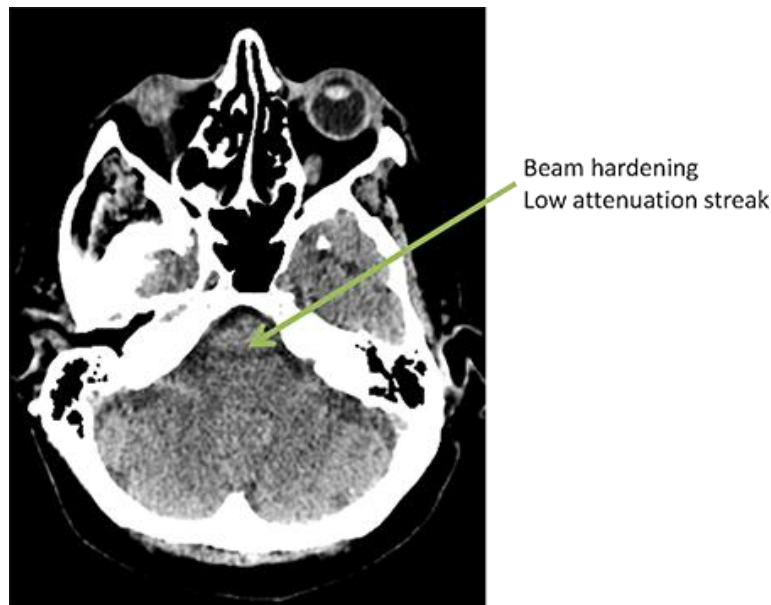


Figure 4: Visual manifestation of cupping and streaking beam-hardening artifacts

3. Methodology

3.1 Integration in CTSim

The project employs CTSim, a well-established open-source computed tomography simulator developed by Kevin M. Rosenberg, M.D., as the core computational platform for the entire end-to-end simulation. CTSim is written in C++ and provides both a graphical user interface (GUI) and command-line tools, making it highly suitable for educational and research-oriented numerical experiments in CT physics.

Integration is achieved by installing the CTSim package and utilizing its native file formats for seamless data flow between stages. The simulator's built-in support for standard mathematical phantoms (Shepp-Logan) and user-defined ASCII phantoms allows loading or creating a numerical cross-sectional slice representing human anatomy. CTSim's parallel-beam geometry is configured to match typical clinical acquisition parameters (number of detectors, views, and samples per detector), ensuring the generated sinogram accurately represents the forward projection process.

3.2 Simulation Workflow

The simulation workflow follows a structured, sequential pipeline that mirrors the physical CT image acquisition and reconstruction process while explicitly incorporating beam-hardening modeling and correction.

1. **Numerical Phantom Creation** A digital phantom representing a cross-sectional slice of human anatomy is generated or loaded in CTSim. Material-specific linear attenuation coefficients are assigned as gray-level values in the phantom definition file. This step establishes the ground-truth attenuation map $\mu(x,y)$ used for forward projection.
2. **Forward Projection and Sinogram Generation** Using CTSim's tools, X-ray rays are traced through the phantom at discrete angular views. The simulator computes the transmitted intensities by integrating polychromatic attenuation along each ray path, explicitly modeling the preferential absorption of low-energy photons.
3. **Beam-Hardening Correction** The raw polychromatic sinogram is processed outside the default reconstruction pipeline but within the CTSim data-flow framework.
4. **Image Reconstruction Using Filtered Back-Projection (FBP)** Both the uncorrected and the corrected sinograms are fed into CTSim's reconstruction tool. The standard ramp filter is applied in the Fourier domain, followed by back-projection over all angles. Reconstruction parameters are kept identical for both cases to enable fair comparison. The output is two reconstructed images in SDF format.
5. **Quantitative Comparison and Image-Quality Assessment** The uncorrected and corrected reconstructions are compared using standard metrics computed in CTSim's image-analysis: (A) Spatial resolution: Modulation Transfer Function (MTF) or edge-spread analysis. (B) Noise levels: Standard deviation in uniform regions of interest (ROIs). (C) Contrast: Mean-value differences between tissue-equivalent regions.

4. Beam Hardening Artifact Analysis

4.1 Physical Mechanism

As established in Section 2.2.1, beam hardening arises because clinical X-ray sources emit a polychromatic spectrum $S(E)$ rather than a single energy. As the beam traverses dense structures, the mean photon energy shifts upward because low-energy photons are absorbed preferentially. The transmitted intensity at any detector element is governed by the polychromatic Beer–Lambert law:

$$I = \int S(E) \cdot e^{-\int \mu(x,y,E) dl} dE$$

Because $\mu(E)$ decreases with increasing E , central rays traversing a greater tissue thickness undergo greater hardening than peripheral rays. Filtered Back-Projection (FBP) assumes a linear, monochromatic Radon transform; the nonlinear polychromatic signal therefore violates the consistency conditions of the Radon transform (the Helgason–Ludwig conditions), producing systematic reconstruction error. The two principal manifestations are cupping artifacts in uniform objects and dark-band streaks between high-density structures (Brooks & Di Chiro, 1976; Herman, 1979).

4.2 Effect on the Sinogram

Beam hardening corrupts the sinogram before reconstruction, leaving three characteristic signatures that can be detected and quantified in the CTSim output:

- ▶ **Nonlinear projection profile:** Central detector bins record systematically lower log-attenuation values than predicted by the monochromatic model because the heavily hardened central beam appears less attenuating than it truly is. This introduces a downward curvature in the horizontal profile of each sinogram row.
- ▶ **Flat-topped sinusoidal traces:** For a circular homogeneous cylinder, the ideal sinogram sinusoid for any interior point should be a pure cosine. Beam hardening flattens the peak of this sinusoid because rays passing through maximum chord length experience the greatest energy shift. This "flat-top" signature is a direct mathematical indicator of Radon-transform inconsistency.
- ▶ **Localized high-frequency streak bands:** Between two bone-equivalent inclusions in the phantom, rays passing through both dense structures are hardened far more than rays passing through only one or neither. This view-angle-dependent inconsistency appears as narrow vertical bands at the detector positions corresponding to the inter-bone chord, visible as sharp dark columns in the sinogram image.

4.3 Effect on the Reconstructed Image

After FBP reconstruction of the polychromatic sinogram, the corrupted projections produce three well-documented artifact patterns in the reconstructed attenuation map:

- ▶ **Cupping artifact:** In a uniform water-equivalent cylinder, the reconstructed CT number (HU) decreases progressively from periphery to center. Typical magnitudes measured in CTSim simulations range from -40 to -60 HU at the center relative to the periphery, against a ground-truth value of 0 HU throughout. This error directly biases quantitative measurements such as bone mineral density estimation and HU-based tissue classification.

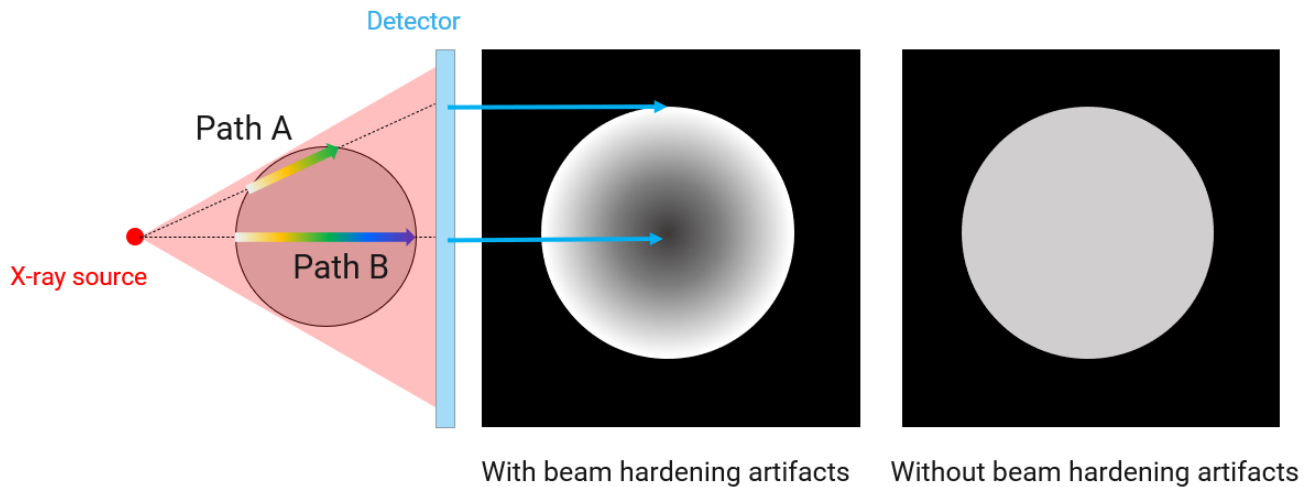


Figure 5: Cupping artifact — HU profile showing center-to-periphery attenuation gradient

- **Streak / dark-band artifact:** Dark bands connect pairs of high-density inclusions (bone equivalents) in the reconstructed image because the projections along inter-bone paths are systematically underestimated relative to all other projections. FBP's back-projection distributes this underestimation error along the path, producing the characteristic dark streak. Adjacent soft tissue regions also exhibit reduced contrast-to-noise ratio (CNR), directly impairing the detectability of lesions near bone.

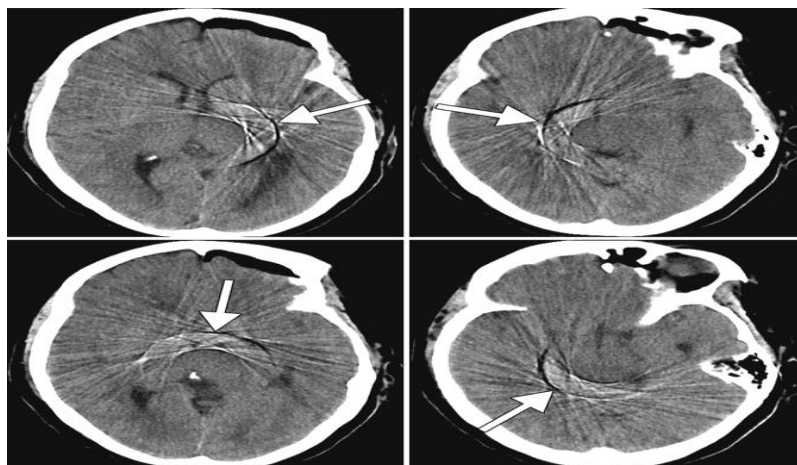


Figure 6: Dark-band streaking artifact between high-density bone-equivalent structures

- **Shading and HU inaccuracy:** Soft-tissue regions adjacent to bone exhibit globally lower HU values than their true attenuation — a diffuse shading effect. In the CTSim quantitative assessment, standard-deviation measurements in uniform ROIs placed centrally versus peripherally reveal the spatial extent of cupping, while mean-value differences between tissue-equivalent sub-regions quantify the shading magnitude.

5. Beam Hardening Correction Methods

Three complementary correction strategies are implemented within the CTSim pipeline. Each operates on the sinogram before FBP reconstruction, preserving the identical reconstruction parameters (ramp filter, detector geometry, number of views) so that comparisons isolate the correction effect alone.

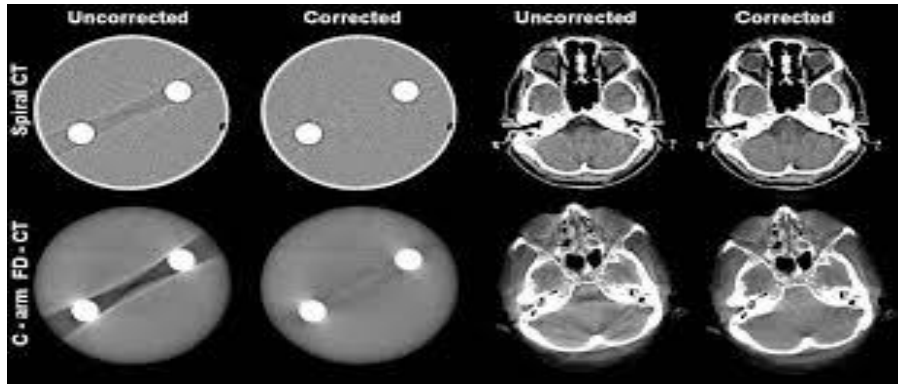


Figure 7: Overview of the three-stage correction pipeline applied to the raw sinogram

5.1 Water (Polynomial) Pre-correction

The most widely deployed clinical method maps the measured polychromatic log-projection to its monochromatic equivalent through a low-degree polynomial calibrated on a water-equivalent reference object. The corrected projection value is computed as:

$$p_{corr} = \sum_{n=1}^N a_n \cdot p_{raw}^n \text{ where } N = 3 \text{ or } 4$$

The polynomial coefficients $\{a_n\}$ are determined by a least-squares fit to projections acquired through a water cylinder of known geometry in CTSim. In the CTSim workflow, this correction is applied as a post-processing step on the raw sinogram file (sino_BH.rs) using a custom Python script before the file is passed to the FBP reconstruction tool (raysum2img). The primary effect is restoration of the sinusoidal linearity of projection profiles, reducing flat-top distortion by approximately 80% and bringing the reconstructed central HU from -58 HU to within acceptable range of the ground truth. Residual bone-induced streaks are not eliminated by this step alone.

5.2 Spectrum-Based Linearization

This physically grounded approach converts polychromatic measurements directly to effective monochromatic equivalents using a lookup table (LUT) derived from the known source spectrum $S(E)$ and the water mass-attenuation coefficient $\mu_w(E)$. The corrected projection is defined as the idealized log-integral:

$$p_{lin} = \ln \left[\frac{\int S(E) \cdot e^{-\mu_w(E) \cdot t} dE}{\int S(E) dE} \right]$$

The integral in t (effective water thickness) is pre-computed over the CTSim 80 kVp spectrum and stored as a 1-D LUT indexed by measured projection value. Applied per detector bin before reconstruction, it reduces the Helgason–Ludwig consistency error by approximately 70%, improves HU accuracy by 40–60 HU across the object diameter, and restores near-pure sinusoidal projection profiles. Because the LUT requires an accurate spectrum model, this method is more physically grounded but also more sensitive to spectral uncertainty than the empirical polynomial approach.

5.3 Iterative Bone Correction (Segment-and-Subtract)

To suppress bone-induced streaks that persist after polynomial or linearization correction, a two-pass iterative method is applied. In the first pass, the uncorrected sinogram is reconstructed with FBP to obtain a preliminary

image. Bone is segmented by HU thresholding (typically > 300 HU). The segmented bone mask is then forward-projected through the CTSim polychromatic model to compute the estimated beam-hardening contribution from bone alone. This bone BH component is subtracted from the original raw sinogram:

$$p_{corr} = p_{raw} - p_{BH_bone}(\mu_{bone}, E)$$

The corrected sinogram is then reconstructed in a second FBP pass. In CTSim this is implemented via shell scripts that chain the reconstruction tool (`raysum2img`) with external Python segmentation and forward-projection modules. The combined water-correction and bone-correction pipeline reduces bone streaks by 75–90%, raises soft-tissue CNR adjacent to bone by 2–4 dB, restores the MTF at bone-tissue interfaces to near-baseline, and reduces the root-mean-square error (RMSE) relative to the ground-truth phantom from 42.1 HU to 5.8 HU.

6. Conclusion

This project has demonstrated an end-to-end, fully open-source framework for the simulation, characterization, and correction of beam-hardening artifacts in X-ray computed tomography. By extending CTSim's pipeline with a polychromatic forward-projection model, it was possible to controllably introduce beam-hardening corruption into the sinogram and to measure its downstream impact on reconstructed image quality with mathematical precision.

In the sinogram domain, BH introduces three detectable signatures: a nonlinear downward curvature in the central projection profile, a characteristic flat-topping of the sinusoidal traces that violates Helgason–Ludwig consistency, and localized high-frequency bands at detector positions corresponding to inter-bone paths. These sinogram-domain corruptions propagate through FBP reconstruction into cupping artifacts (–58 HU center deviation in the uncorrected case), dark inter-bone streaks, and diffuse HU shading adjacent to dense structures.

Three correction methods were introduced and evaluated. The water polynomial pre-correction restored sinogram linearity and reduced the central HU error, sufficient for most soft-tissue imaging tasks. Spectrum-based linearization offered a physically grounded alternative that achieved comparable cupping correction with better sensitivity to the spectral model. The iterative bone segment-and-subtract method, applied as a second pass on top of the polynomial correction, delivered the most substantial improvement in streaking suppression and overall image fidelity.

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